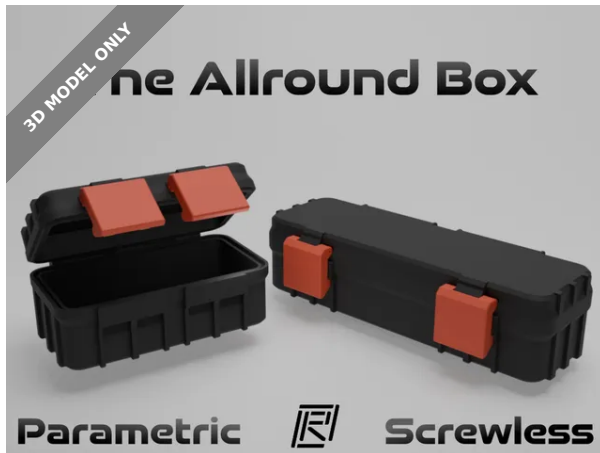




Rugged Storage Box (Parametric, Screwless)



RDesigns

[VIEW IN BROWSER](#)

updated 14. 12. 2024 | published 14. 12. 2024

Summary

This is a rugged storage box. Keyfeatures: - Screwless design - Easy to customize - Parametric design

[Hobby & Makers](#) > [Organizers](#)

Tags: [parametric](#) [box](#) [organizer](#) [storage](#) [customizable](#)
[easy](#) [organization](#) [screwless](#) [rugged](#) [ruggedbox](#)

This is a Screwless, parametric, sealed, storage box.

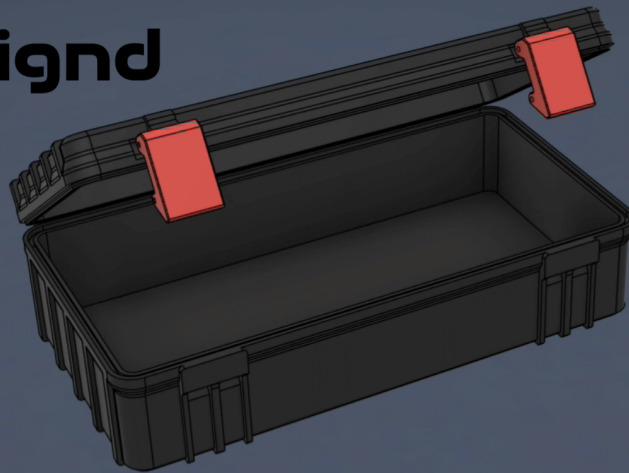
The Alround Box MINI: <https://www.printables.com/de/model/851837-rugged-storage-box-mini-parametric-screwless>



Rugged Box Design

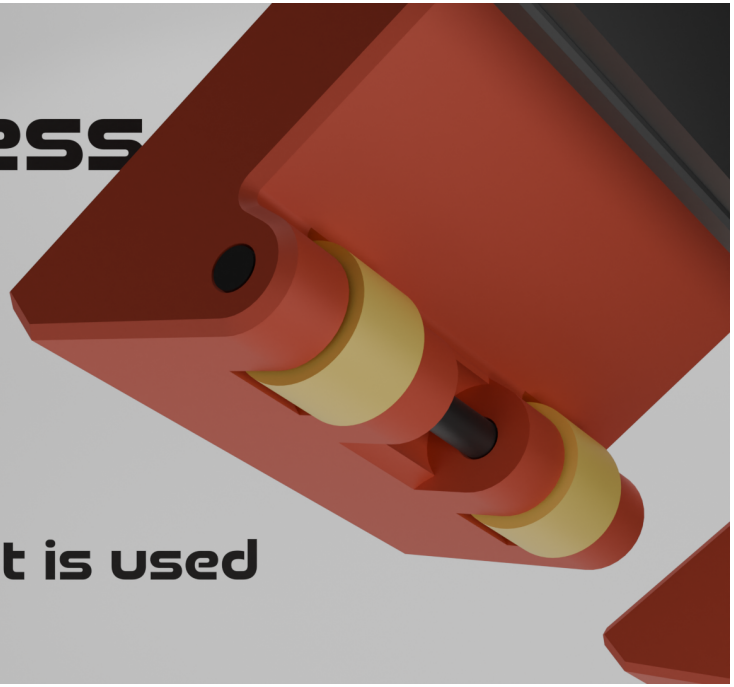


Parametric Design



Screwless Design

100% filament is used



Sealed with
TPU filament

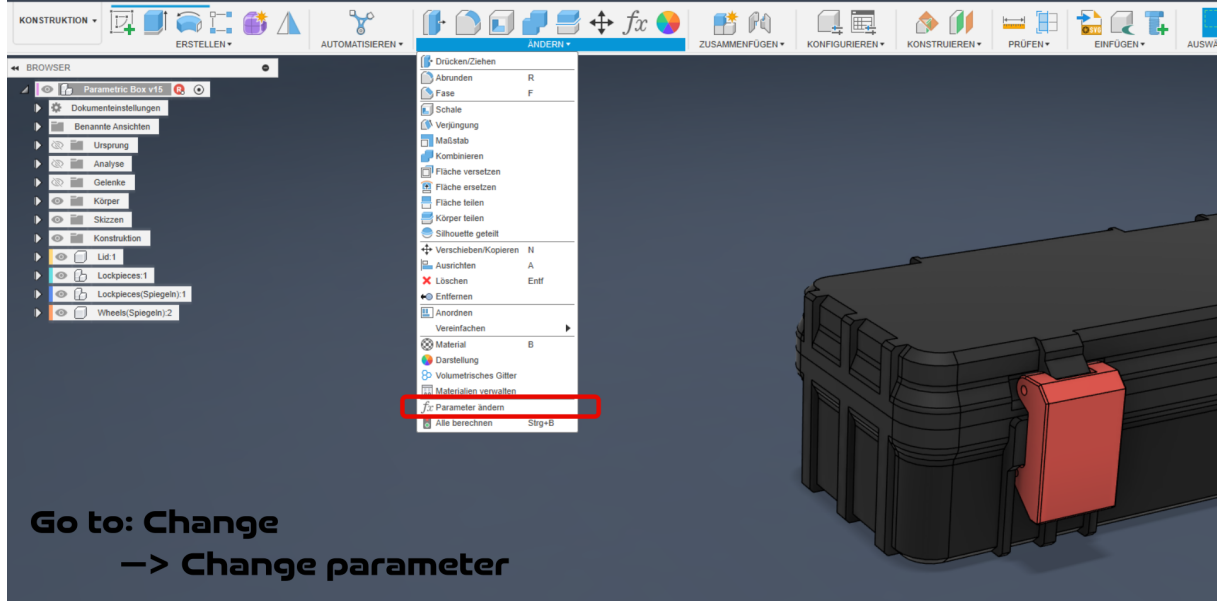


If no sealing is used,
please make sure to use
special wheels file



Guides

How to easily change the parameters:



Parametric designed in Fusion 360

Alle Parameter filtern

Parameter	Name	Einheit	Ausdruck	Wert	Kommentare
★ Favoriten					
★ Benutzerparam...	Insidewidth	mm	50 mm	50.00	Minimum: 20
★ Benutzerparam...	InsideHeight	mm	40 mm	40.00	Minimum: 30
★ Benutzerparam...	InsideLength	mm	200 mm	200.00	Minimum: ca. 100 (Adjust CornerRadius to go further below 100)
★ Benutzerparam...	CornerRadius	mm	8 mm	8.00	Minimum: 5 (Also adjusts the latch position)
★ Benutzerparam...	WallThickness	mm	3.5 mm	3.50	If not necessary, don't Change
★ Benutzerparam...	HingeThickness	mm	25 mm	25.00	Minimum: 10
★ Benutzerparam...	LockWidth	mm	20 mm	20.00	Minimum: 18
★ Benutzerparam...	RibDistance	mm	22 mm	22.00	Change if one of the Ribs is lost

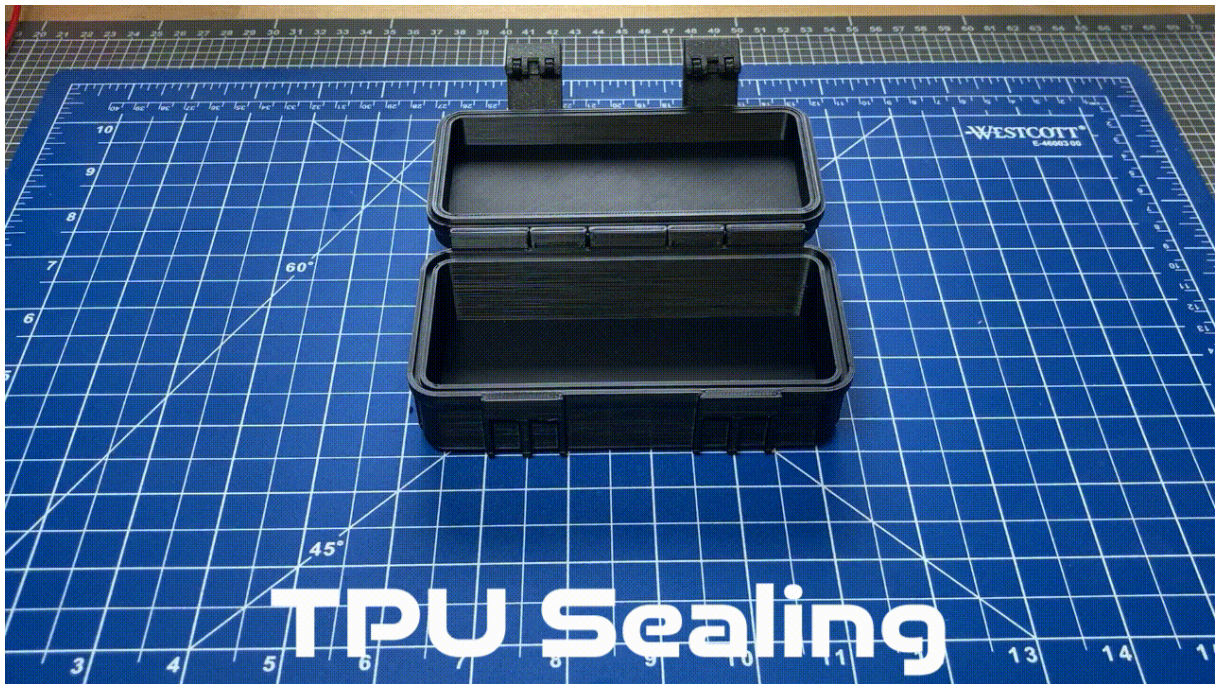
Change these values

Attention to the comments

OK

Assembly





The box is assembled with pieces of filament, so screws aren't needed anymore!

For the sealing, TPU can be used.

Attention:

Pay attention to the comments in the parametric design in Fusion 360!

Release Notes:

27.03.2024

- The V1 available now!
 - Screws are replaced through small pieces of Filament that perfectly fit in!
 - The Design in Fusion 360 is parametric, most of the dimension can be changed.
 - The latch design stands out with its wheels, which lock into place perfectly!

31.03.2024

- Update V1.2!
 - Some model issues fixed.
 - Better fit for non sealed boxes.
 - Filament holes widened.

29.06.2024

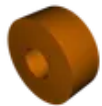
- Update V1.5!
 - Now you can change the latch size in the parametric fusion 360 file.
 - New latch option for better user experience without the TPU seal (parametric f3d)
 - New 3d modell renderings.

Model files



Latch and Wheels

2 files



wheel.3mf



latch.3mf



Parametric Fusion 360 file

1 file

parametric-box-v152.f3d

☐ Fusion update problem fixed



Box 180x200x100

2 files



lid-180x200x100.3mf



mainbody-180x200x100.3mf



Box 100x50x40

2 files



mainbody-100x50x40.3mf



lid-100x50x40.3mf



Box 100x100x40

2 files



lid-100x100x40.3mf



mainbody-100x100x40.3mf



Box 160x120x40

2 files



mainbody-160x120x40.3mf



lid-160x120x40.3mf



Box 160x120x60

2 files



mainbody-160x120x60.3mf



lid-160x120x60.3mf



Box 160x80x60

2 files



mainbody-160x80x60.3mf



lid-160x80x60.3mf



Box 160x50x60

2 files



lid-160x50x60.3mf



mainbody-160x50x60.3mf



Box 120x50x60

2 files



mainbody-120x50x60.3mf



mainbody-120x80x60.3mf



Box 120x80x60

2 files



lid-120x50x60.3mf



lid-120x80x60.3mf



Box 100x80x40

2 files



lid-100x80x40.3mf



mainbody-100x80x40.3mf



Box 100x50x60

2 files



mainbody-100x50x60.3mf



lid-100x50x60.3mf



Box 100x80x60

2 files



mainbody-100x80x60.3mf



lid-100x80x60.3mf



Box 160x80x40

2 files



mainbody-160x80x40.3mf



lid-160x80x40.3mf



Box 160x50x40

2 files



mainbody-160x50x40.3mf



lid-160x50x40.3mf



Box 120x120x40

2 files



lid-120x120x40.3mf



mainbody-120x120x40.3mf



Box 120x80x40

2 files



lid-120x80x40.3mf



mainbody-120x80x40.3mf



Box 120x50x40

2 files



mainbody-120x50x40.3mf



lid-120x50x40.3mf

License

This work is licensed under a
[Creative Commons \(4.0 International License\)](https://creativecommons.org/licenses/by/4.0/)



Attribution

-
- ✗ | Sharing without ATTRIBUTION
 - ✓ | Remix Culture allowed
 - ✓ | Commercial Use
 - ✓ | Free Cultural Works
 - ✓ | Meets Open Definition